AXEL v2 — Phase 6 Product Requirements Document

Purpose

Phase 6 is the first controlled transition of the Stocks section from paper operation to real capital. It does not introduce new strategy intelligence. Its purpose is to prove that the already-validated Stocks system can operate with real orders under strict deterministic controls, tiny allocation limits, manual approval requirements and immediate halt/reconciliation capability. Phase 6 is blocked until the Phase 5 Stocks paper-soak promotion gate has passed with its required evidence. No developer may bypass the gate by configuration.

Required Scope

The phase implements live configuration gating, broker integration, manual approval, tiny-live risk limits, order lifecycle management, live reconciliation, incident monitoring, kill/stop controls and rollback/recovery. It must preserve the AXEL rule that LLM/agents propose while the deterministic Trading core validates and the independent Watchdog can halt activity.

Goals

• Connect the approved Stocks execution path to a supported broker. • Prove order lifecycle correctness from proposal through fill. • Enforce live-mode gates and manual approval. • Enforce tiny capital, position, loss and order limits. • Reconcile broker state against AXEL state continuously. • Detect incidents and halt safely. • Prove rollback and recovery procedures. • Produce an auditable record of every live action.

Non-Goals

No new Stocks strategy development, autonomous scaling, removal of manual approval, risk-limit relaxation, multi-broker expansion, or activation of Crypto/Forex/Options/Bets occurs here.

Preconditions

• Phase 5 promotion gate is passed and evidence recorded. • Stocks strategy/playbook versions are frozen. • Risk configuration is reviewed and versioned. • Broker account is explicitly designated for AXEL tiny-live use. • API permissions are least privilege. • Kill-switch and Watchdog drills pass. • Reconciliation tests pass. • Manual approval workflow passes. • Rollback/halt runbook is approved. • Operators can access logs, alerts and emergency controls.

Live Configuration Gate

Live mode must require an explicit versioned configuration containing environment=LIVE, approved broker/account, allowed section=STOCKS, maximum live capital, maximum position notional, maximum order notional, maximum daily loss, maximum concurrent positions, maximum order frequency, allowed universe, manual-approval requirement, kill-switch state and configuration approval metadata. Missing, stale, conflicting or invalid values fail closed.

Tiny-Live Risk Limits

The initial live envelope must be materially smaller than the validated/paper portfolio. Exact numerical values are deployment configuration and must be separately approved. The implementation must support hard ceilings for total exposure, per-position exposure, daily loss, order size, order frequency and concentration. Invalid orders are rejected before broker submission.

Broker Adapter

Expose a deterministic interface equivalent to: Provider-specific responses remain inside the adapter. Broker credentials are available only to the execution process.

Order State Machine

Orders use explicit states such as \[CREATED APPROVED SUBMITTED ACKNOWLEDGED PARTIALLY_FILLED FILLED\] with CANCELLED, REJECTED, EXPIRED and ERROR paths. Every transition is validated. Duplicate submission is prevented using idempotency keys and deterministic client order IDs.

Manual Approval

Approval must be an explicit authenticated action tied to the exact proposal/order hash and configuration version. It expires after a defined window and cannot be reused for a materially different order. Proposal creation never implies approval.

Pre-Trade Validation

• Validate schema and strategy/playbook eligibility. • Verify market/session and symbol eligibility. • Calculate deterministic risk and check exposure/concentration. • Check daily loss/drawdown, duplicate/stale order and kill-switch. • Verify Watchdog health. • Verify manual approval where required. • Write immutable audit record. • Submit only after all checks pass.

Post-Trade Reconciliation

Continuously compare AXEL state with broker open orders, fills, positions, cash/buying power, realized/unrealized P&L, cancellations and rejections. Any unexplained mismatch becomes an incident; discrepancies are never silently overwritten.

Incident Handling

Incidents include unexpected fills, unknown orders, position mismatch, stale market data, broker outage, repeated rejects, excessive loss, duplicate submission, Watchdog failure, database failure and authentication failure. Emergency conditions invoke the deterministic stop path.

Kill-Switch and Halt

Support global halt, Stocks-section halt, new-order halt while reconciliation/cancellation remains available, broker-disconnect halt, risk-limit halt and Watchdog-triggered halt. Halt behavior must be deterministic and tested under failure.

Rollback and Recovery

The runbook must define stopping new orders, inspecting/cancelling outstanding orders, broker reconciliation, position verification, incident recording, return to paper mode, recovery authority and evidence required before re-enabling live. Recovery never silently restores live permissions.

Observability

Expose order lifecycle, broker connectivity, open orders, positions, exposure, P&L, risk checks, rejected orders, reconciliation status, Watchdog heartbeat, kill-switch state and incident IDs. Every live action has a correlation ID.

Security

Broker secrets use approved secret management, never appear in logs/prompts/business records, and are inaccessible to agents/research. Live DB roles are least privilege. Production configuration changes are reviewed and audited.

Testing

Contract Tests

Broker order creation, cancellation, status, fills, rejections and account/position retrieval.

Failure Tests

Timeout, duplicate response, delayed/partial fill, rejection, network loss, database outage, stale data, Watchdog failure and kill-switch activation.

Reconciliation Tests

Intentional mismatches must be detected and escalated.

Permission Tests

Agents cannot submit orders; live mode cannot activate without valid configuration; approvals cannot be forged or reused.

Live Rollout Sequence

• Keep live configuration disabled. • Run broker connectivity and read-only account checks. • Run sandbox order lifecycle tests. • Enable tiny-live permissions. • Require manual approval for every order initially. • Execute the smallest approved order. • Verify fill and reconciliation. • Continue only inside the tiny limits. • Review evidence before any future increase.

Assignable Work Packages

Developer Boundaries

Developers must not remove manual approval, weaken risk limits, disable Watchdog/kill-switch controls, expose broker credentials, or create a direct Agent-to-broker path.

Definition of Done

• Phase 5 evidence is verified. • Live configuration fails closed when incomplete. • Broker adapter passes contract tests. • Order FSM handles expected states and duplicate submission is prevented. • Approval is bound to the exact order. • Pre-trade gates reject invalid orders. • Reconciliation detects intentional mismatches. • Kill/halt drills pass. • Incident and rollback runbooks are tested. • Initial tiny-live operation is fully auditable.

Acceptance Test

From a clean deployment with live disabled, an operator cannot submit an order. After approved configuration and manual approval, the smallest Stocks order can pass deterministic validation and reach the broker. Its fill is reconciled. A simulated mismatch creates an incident. A kill-switch event stops new orders. Recovery returns the system to disabled/paper state until explicitly reauthorized.

Handoff

Phase 6 establishes a proven live execution boundary for Stocks. Phase 7 Crypto must reuse the deterministic core and Watchdog while adding its own data, strategies and section-specific controls.